Project Report

1. PROJECT TITLE

Predictive Pulse: Harnessing Machine Learning for Blood Pressure Analysis

2. PROJECT DESCRIPTION

System Overview

Predictive Pulse is a machine learning based healthcare system designed to analyze patient health data and predict blood pressure levels. The system uses machine learning algorithms to identify patterns in health parameters and estimate the risk of hypertension.

Problem Domain

Hypertension is a major health problem worldwide. Many individuals are unaware of their blood pressure condition until it leads to serious diseases such as heart attacks or strokes. Early prediction can help in preventing these complications.

Core Technology Used

- Python
- Machine Learning Algorithms
- Data Analysis Libraries (Pandas, NumPy)
- Visualization Tools (Matplotlib, Seaborn)

Key Capabilities

- Health data analysis
- Blood pressure prediction
- Data visualization
- Risk identification

Target Users

- Healthcare professionals
- Patients
- Medical researchers
- Health monitoring systems

Real-world Relevance

The system helps in early detection of hypertension and assists healthcare providers in making better decisions based on patient data.

3. APPLICATION SCENARIOS / USE CASES

- **Hospital Usage**

Doctors can use the system to analyze patient health data and predict potential blood pressure problems.

- **Personal Health Monitoring**

Individuals can use the system to monitor their health and check hypertension risk.

- **Research and Medical Studies**

Researchers can analyze health datasets to study patterns related to blood pressure.

- **Telemedicine Platforms**

The system can be integrated into online healthcare platforms to assist remote diagnosis.

4. TECHNICAL ARCHITECTURE OVERVIEW

Frontend Layer

User interface where users input health information such as age, BMI, heart rate, etc.

Backend Layer

The backend processes the input data using machine learning models to generate predictions.

Database Layer

Stores the dataset and prediction results.

External Services

Machine learning libraries and data analysis tools.

Deployment Environment

The system can be deployed using local servers, cloud platforms, or web applications

5. PREREQUISITES

5.1 Software Requirements

- Python 3.x
- Jupyter Notebook / Google Colab
- VS Code or PyCharm
- Anaconda (optional)

5.2 Libraries / Frameworks / Dependencies

- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

5.3 Hardware Requirements

- Minimum 4GB RAM
- Intel i3 processor or higher
- Internet connection (for dataset download)

6. PRIOR KNOWLEDGE REQUIRED

- Basic Python programming
- Machine learning fundamentals
- Data analysis concepts
- Understanding of healthcare datasets
- Knowledge of Python libraries such as Pandas and Scikit-learn

7. PROJECT OBJECTIVES

Technical Objectives

- Build a machine learning model to predict blood pressure.

Performance Objectives

- Achieve high accuracy in prediction.

Deployment Objectives

- Implement the system in a user-friendly environment.

Learning Outcomes

- Understanding machine learning in healthcare.
- Data analysis and model development.

8. SYSTEM WORKFLOW

1. User enters health parameters.
2. System collects input data.
3. Data preprocessing is performed.
4. Machine learning model processes the data.
5. Prediction is generated.
6. Result is displayed to the user.

9. MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN

9.1 Problem Definition

Early detection of hypertension is difficult without regular medical tests. A predictive system can help analyze health data and identify risks.

9.2 Functional Requirements

- Input patient health data
- Process and analyze data
- Predict blood pressure level
- Display results to users

9.3 Non-Functional Requirements

- High system accuracy
- Fast processing time
- Easy user interface

9.4 System Design Decisions

- Use machine learning models for prediction.
- Use Python libraries for data processing and analysis.

9.5 Technology Stack Selection Justification

Python is widely used in machine learning due to its powerful libraries and simplicity.

10. MILESTONE 2: ENVIRONMENT SETUP & INITIAL CONFIGURATION

10.1 Development Environment Setup

Install Python and development tools such as Jupyter Notebook.

10.2 Dependency Installation

Example command:

pip install pandas numpy scikit-learn matplotlib seaborn

10.3 Project Structure Creation

Example folder structure:

```
PredictivePulse/
│
├── dataset/
├── models/
├── notebooks/
├── src/
├── outputs/
└── requirements.txt
```

10.4 Configuration Setup

Configure dataset paths and model parameters in the code.

11. MILESTONE 3: CORE SYSTEM DEVELOPMENT

Feature 1 – Data Preprocessing

Cleaning and preparing the dataset for analysis.

Feature 2 – Model Training

Training machine learning models such as:

- Logistic Regression
- Decision Tree
- Random Forest

Feature 3 – Prediction System

The trained model predicts blood pressure levels based on input data.

12. MILESTONE 4: INTEGRATION & OPTIMIZATION

Component Integration

Combine dataset processing, model training, and prediction modules.

Performance Optimization

Optimize model parameters to improve accuracy.

Security Enhancements

Protect sensitive health data.

Error Handling & Validation

Handle invalid inputs and missing data.

13. MILESTONE 5: TESTING & VALIDATION

13.1 Test Cases

Test Case ID	Input	Expected Output	Status
TC01	Valid health data	BP prediction	Pass
TC02	Missing values	Error message	Pass
TC03	Extreme values	Prediction result	Pass

13.5 Performance Metrics

- Accuracy
- Precision
- Recall

- F1 Score

Example accuracy: **85% – 92%**

14. DEPLOYMENT

Deployment Architecture

The model can be deployed as a web application or integrated into healthcare systems.

Hosting Platform

- Local server
- Cloud platforms (AWS, Azure)

Deployment Steps

1. Train the machine learning model.
2. Save the model.
3. Connect the model with user interface.

Production Considerations

Ensure system security and data privacy.

16. RESULTS

- The model successfully predicts blood pressure risk.
- Random Forest achieved the highest accuracy.
- The system demonstrates reliable prediction results.

17. ADVANTAGES & LIMITATIONS

Advantages

- Early detection of hypertension
- Fast prediction results
- Helps doctors in decision-making

Limitations

- Accuracy depends on dataset quality
- Requires large medical datasets

- Cannot replace medical diagnosis

18. FUTURE ENHANCEMENTS

- Integration with wearable devices
- Mobile application development
- Real-time health monitoring
- Deep learning model implementation
- Cloud-based healthcare system

19. CONCLUSION

The Predictive Pulse system demonstrates the use of machine learning techniques in healthcare to predict blood pressure levels. By analyzing patient health data, the system helps detect hypertension risks early. This project highlights the potential of artificial intelligence in improving healthcare systems and supporting medical professionals.

20. APPENDIX

- **Source code**

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hypertension Risk Assessment - AI Platform</title>

  <!-- Google Fonts -->
  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display
=swap" rel="stylesheet">
```

```
<!-- Tailwind CSS CDN -->
<script src="https://cdn.tailwindcss.com"></script>

<!-- Lucide Icons -->
<script src="https://unpkg.com/lucide@latest"></script>

<style>
    /* Custom CSS */
    body {
        font-family: 'Inter', sans-serif;
        background: linear-gradient(135deg, #4f46e5 0%, #7c3aed 50%, #9333ea 100%);
        min-height: 100vh;
    }

    .glass-card {
        background: rgba(255, 255, 255, 0.95);
        backdrop-filter: blur(8px);
        border: 1px solid rgba(255, 255, 255, 0.2);
        box-shadow: 0 20px 25px -5px rgba(0, 0, 0, 0.1);
        border-radius: 1.5rem;
    }

    .input-field {
        width: 100%;
        padding: 0.75rem 1rem;
        background-color: #f8fafc;
        border: 1px solid #e2e8f0;
        border-radius: 0.75rem;
```

```
    font-size: 0.875rem;
    transition: all 0.2s;
    outline: none;
    appearance: none;
    background-image: url("data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg' fill='none' viewBox='0 0 24 24' stroke='%2364748b'%3E%3Cpath stroke-linecap='round' stroke-linejoin='round' stroke-width='2' d='M19 9l-7 7-7-7'%3E%3C/path%3E%3C/svg%3E");
    background-repeat: no-repeat;
    background-position: right 0.75rem center;
    background-size: 1rem;
  }

  .input-field:focus {
    border-color: #6366f1;
    box-shadow: 0 0 0 3px rgba(99, 102, 241, 0.2);
  }

  .label-text {
    display: flex;
    align-items: center;
    gap: 0.5rem;
    font-size: 0.75rem;
    font-weight: 700;
    color: #475569;
    text-transform: uppercase;
    letter-spacing: 0.05em;
    margin-bottom: 0.5rem;
  }
```

```
        /* Modal Styles */
        .modal-overlay {
            position: fixed;
            inset: 0;
            z-index: 100;
            display: flex;
            align-items: center;
            justify-content: center;
            padding: 1rem;
            background: rgba(15, 23, 42, 0.7);
            backdrop-filter: blur(4px);
        }

        .custom-scrollbar::-webkit-scrollbar { width: 6px; }
        .custom-scrollbar::-webkit-scrollbar-track { background: #f8fafc; }
        .custom-scrollbar::-webkit-scrollbar-thumb { background: #cbd5e1; border-radius: 10px;
}
    </style>
</head>
<body class="antialiased text-slate-900 p-4 md:p-8">

    <div class="max-w-6xl mx-auto">
        <!-- Header -->
        <header class="flex flex-col md:flex-row items-center justify-between mb-10 gap-6">
            <div class="flex items-center gap-4">
                <div class="w-14 h-14 bg-white rounded-2xl shadow-lg flex items-center justify-
center">
                    <i data-lucide="activity" class="w-8 h-8 text-indigo-600"></i>
                </div>
                <div>
```

```
            <h1 class="text-3xl font-bold text-white tracking-tight">Hypertension Detection</h1>
            <p class="text-indigo-100 text-sm opacity-90">AI-Powered Cardiovascular Risk Assessment</p>
          </div>
        </div>
        <div class="flex gap-2">
          <span class="px-3 py-1 bg-white/20 border border-white/30 rounded-full text-white text-[10px] font-bold uppercase tracking-wider flex items-center gap-1.5">
            <i data-lucide="shield-check" class="w-3 h-3"></i> Medical Grade
          </span>
          <span class="px-3 py-1 bg-white/20 border border-white/30 rounded-full text-white text-[10px] font-bold uppercase tracking-wider flex items-center gap-1.5">
            <i data-lucide="zap" class="w-3 h-3"></i> AI-Powered
          </span>
        </div>
      </header>

      <main class="grid grid-cols-1 lg:grid-cols-12 gap-8">
        <!-- Form Section -->
        <div class="lg:col-span-8">
          <form action="/predict" method="POST" class="glass-card p-8">
            <div class="flex items-center gap-3 mb-8">
              <i data-lucide="stethoscope" class="w-6 h-6 text-indigo-600"></i>
              <h2 class="text-xl font-bold text-slate-800">Comprehensive Health Assessment</h2>
            </div>

            <div class="grid grid-cols-1 md:grid-cols-2 gap-6">
              <!-- 1. Gender -->
              <div>
```

```
                <label class="label-text"><i data-lucide="user" class="w-3.5 h-3.5"></i>
Gender</label>
                <select name="gender" class="input-field">
                  <option value="Male">Male</option>
                  <option value="Female">Female</option>
                </select>
              </div>

              <!-- 2. Age Group -->
              <div>
                <label class="label-text"><i data-lucide="calendar" class="w-3.5 h-3.5"></i>
Age Group</label>
                <select name="ageGroup" class="input-field">
                  <option value="18-34 years">18-34 years</option>
                  <option value="35-44 years">35-44 years</option>
                  <option value="45-54 years">45-54 years</option>
                  <option value="55-64 years">55-64 years</option>
                  <option value="65+ years">65+ years</option>
                </select>
              </div>

              <!-- 3. Family History -->
              <div>
                <label class="label-text"><i data-lucide="history" class="w-3.5 h-3.5"></i>
Family History</label>
                <select name="familyHistory" class="input-field">
                  <option value="No">No</option>
                  <option value="Yes">Yes</option>
                </select>
              </div>
```

```
<!-- 4. Medical Care -->
<div>
  <label class="label-text"><i data-lucide="heart-pulse" class="w-3.5 h-3.5"></i> Under Medical Care</label>
  <select name="medicalCare" class="input-field">
    <option value="No">No</option>
    <option value="Yes">Yes</option>
  </select>
</div>

<!-- 5. BP Medication -->
<div>
  <label class="label-text"><i data-lucide="pill" class="w-3.5 h-3.5"></i> Taking BP Medication</label>
  <select name="bpMedication" class="input-field">
    <option value="No">No</option>
    <option value="Yes">Yes</option>
  </select>
</div>

<!-- 6. Symptom Severity -->
<div>
  <label class="label-text"><i data-lucide="alert-triangle" class="w-3.5 h-3.5"></i> Symptom Severity</label>
  <select name="symptomSeverity" class="input-field">
    <option value="None">None</option>
    <option value="Mild">Mild</option>
    <option value="Moderate">Moderate</option>
    <option value="Severe">Severe</option>
```

```
            </select>
        </div>

        <!-- 7. Shortness of Breath -->
        <div>
            <label class="label-text"><i data-lucide="wind" class="w-3.5 h-3.5"></i>
Shortness of Breath</label>
            <select name="shortnessOfBreath" class="input-field">
                <option value="No">No</option>
                <option value="Yes">Yes</option>
            </select>
        </div>

        <!-- 8. Vision Changes -->
        <div>
            <label class="label-text"><i data-lucide="eye" class="w-3.5 h-3.5"></i>
Vision Changes</label>
            <select name="visionChanges" class="input-field">
                <option value="No">No</option>
                <option value="Yes">Yes</option>
            </select>
        </div>

        <!-- 9. Nosebleeds -->
        <div>
            <label class="label-text"><i data-lucide="droplets" class="w-3.5 h-3.5"></i>
Frequent Nosebleeds</label>
            <select name="nosebleeds" class="input-field">
                <option value="No">No</option>
                <option value="Yes">Yes</option>
```

```
            </select>
        </div>

        <!-- 10. Time Since Diagnosis -->
        <div>
            <label class="label-text"><i data-lucide="clock" class="w-3.5 h-3.5"></i>
Time Since Diagnosis</label>
            <select name="timeSinceDiagnosis" class="input-field">
                <option value="N/A">N/A</option>
                <option value="Less than 1 Year">Less than 1 Year</option>
                <option value="1-5 Years">1-5 Years</option>
                <option value="5+ Years">5+ Years</option>
            </select>
        </div>

        <!-- 11. Systolic -->
        <div>
            <label class="label-text text-rose-600"><i data-lucide="arrow-up" class="w-
3.5 h-3.5"></i> Systolic Pressure</label>
            <select name="systolic" class="input-field">
                <option value="Less than 120 mmHg (Normal)">Less than 120 mmHg
(Normal)</option>
                <option value="120-129 mmHg (Elevated)">120-129 mmHg
(Elevated)</option>
                <option value="130-139 mmHg (Stage 1)">130-139 mmHg (Stage
1)</option>
                <option value="140-180 mmHg (Stage 2)">140-180 mmHg (Stage
2)</option>
                <option value="Higher than 180 mmHg (Crisis)">Higher than 180 mmHg
(Crisis)</option>
            </select>
        </div>
```

```
<!-- 12. Diastolic -->
<div>
  <label class="label-text text-indigo-600"><i data-lucide="arrow-down" class="w-3.5 h-3.5"></i> Diastolic Pressure</label>
  <select name="diastolic" class="input-field">
    <option value="Less than 80 mmHg (Normal)">Less than 80 mmHg (Normal)</option>
    <option value="80-89 mmHg (Stage 1)">80-89 mmHg (Stage 1)</option>
    <option value="90-120 mmHg (Stage 2)">90-120 mmHg (Stage 2)</option>
    <option value="Higher than 120 mmHg (Crisis)">Higher than 120 mmHg (Crisis)</option>
  </select>
</div>

<!-- 13. Diet -->
<div class="md:col-span-2">
  <label class="label-text"><i data-lucide="utensils" class="w-3.5 h-3.5"></i> Heart-Healthy Diet</label>
  <select name="diet" class="input-field">
    <option value="Yes">Yes</option>
    <option value="No">No</option>
    <option value="Sometimes">Sometimes</option>
  </select>
</div>
</div>

<button type="submit" class="w-full mt-10 py-4 bg-indigo-600 hover:bg-indigo-700 text-white font-bold rounded-xl shadow-lg shadow-indigo-200 transition-all flex items-center justify-center gap-3">
  <i data-lucide="activity" class="w-5 h-5"></i> GENERATE RISK ASSESSMENT
```

```
            </button>
        </form>
    </div>

    <!-- Sidebar -->
    <aside class="lg:col-span-4 space-y-6">
        <section class="glass-card p-6">
            <h3 class="text-lg font-bold text-slate-800 flex items-center gap-2 mb-4">
                <i data-lucide="info" class="w-5 h-5 text-indigo-600"></i> About System
            </h3>
            <p class="text-slate-600 text-xs leading-relaxed mb-4">
                AI-powered platform analyzing cardiovascular risk factors using advanced
machine learning algorithms.
            </p>
            <ul class="space-y-2 text-[11px] text-slate-500 font-medium">
                <li class="flex items-center gap-2"><div class="w-1 h-1 bg-indigo-400 rounded-
full"></div> Evidence-based assessment</li>
                <li class="flex items-center gap-2"><div class="w-1 h-1 bg-indigo-400 rounded-
full"></div> Clinical guideline compliance</li>
                <li class="flex items-center gap-2"><div class="w-1 h-1 bg-indigo-400 rounded-
full"></div> Personalized recommendations</li>
                <li class="flex items-center gap-2"><div class="w-1 h-1 bg-indigo-400 rounded-
full"></div> Professional Medical Support</li>
            </ul>
        </section>

        <section class="glass-card p-6">
            <h3 class="text-lg font-bold text-slate-800 flex items-center gap-2 mb-4">
                <i data-lucide="heart" class="w-5 h-5 text-rose-500"></i> BP Guidelines
            </h3>
            <div class="space-y-2 text-[11px]">
```

```
                <div class="flex justify-between border-b border-slate-100 pb-1">
                  <span class="font-bold">Normal:</span> <span class="text-slate-500">&lt;120/80 mmHg</span>
                </div>
                <div class="flex justify-between border-b border-slate-100 pb-1">
                  <span class="font-bold">Elevated:</span> <span class="text-slate-500">120-129/&lt;80 mmHg</span>
                </div>
                <div class="flex justify-between border-b border-slate-100 pb-1">
                  <span class="font-bold">Stage 1:</span> <span class="text-slate-500">130-139/80-89 mmHg</span>
                </div>
                <div class="flex justify-between border-b border-slate-100 pb-1">
                  <span class="font-bold">Stage 2:</span> <span class="text-slate-500">140-180/90-120 mmHg</span>
                </div>
                <div class="flex justify-between">
                  <span class="font-bold">Crisis:</span> <span class="text-rose-600 font-bold">&gt;180/&gt;120 mmHg</span>
                </div>
              </div>
            </section>
          </aside>
        </main>

        <footer class="mt-12 py-6 border-t border-white/10 text-center">
          <p class="text-white/80 text-sm font-medium">Project made by Sri Balaji University Pune (SCS)</p>
          <p class="text-white/60 text-xs mt-1">Hrushikesh Takmoge: MCA2504133</p>
          <p class="text-white/60 text-xs mt-1">Joey Herbert: MCA2504167</p>
          <p class="text-white/60 text-xs mt-1">Abhijeet Gaikwad: MCA2504081</p>
```

```
        <p class="text-white/60 text-xs mt-1">Gaurav Kautkar: MCA2504086</p>
      </footer>
    </div>

    <!-- Result Modal (Flask Logic) -->
    {% if prediction_text %}
    <div class="modal-overlay" id="resultModal">
      <div class="bg-white w-full max-w-2xl rounded-3xl shadow-2xl overflow-hidden relative flex flex-col max-h-[90vh]">
        <!-- Modal Header -->
        <div class="p-6 border-b border-slate-100 flex items-center justify-between bg-slate-50/50">
          <div class="flex items-center gap-3">
            <div class="w-10 h-10 bg-indigo-600 rounded-xl flex items-center justify-center shadow-lg">
              <i data-lucide="activity" class="w-6 h-6 text-white"></i>
            </div>
            <h3 class="text-xl font-bold text-slate-800">Assessment Result</h3>
          </div>
          <button onclick="closeModal()" class="p-2 hover:bg-slate-200 rounded-full transition-colors">
            <i data-lucide="x" class="w-6 h-6 text-slate-400"></i>
          </button>
        </div>

        <!-- Modal Content -->
        <div class="p-8 overflow-y-auto custom-scrollbar">
          <div
            class="mb-8 p-6 border-2 border-l-8 rounded-2xl"
            id="resultCard"
            data-result-color="{{ result_color }}"
```

```
>
    <div class="flex items-start gap-4 mb-6">
        <div class="w-12 h-12 rounded-full flex items-center justify-center shrink-0 shadow-lg" id="resultIconBg">
            <i data-lucide="heart" class="w-7 h-7 text-white"></i>
        </div>
        <div class="flex-1">
            <div class="flex items-center justify-between mb-1">
                <h3 class="text-xl font-bold" id="resultTitle">{{ prediction_text }}</h3>
                <span class="px-3 py-1 rounded-full text-[10px] font-bold text-white" id="resultPriority">{{ recommendation.priority }}</span>
            </div>
            <p class="text-xs font-medium text-slate-500 mb-2">AI Confidence: {{ confidence }}%</p>
            <p class="text-sm text-slate-600 leading-relaxed">{{ recommendation.description }}</p>
        </div>
    </div>


    <div class="space-y-3">
        <h4 class="text-sm font-bold text-slate-700 flex items-center gap-2 mb-4">
            <i data-lucide="check-circle" class="w-4 h-4 text-indigo-600"></i> Clinical Recommendations
        </h4>
        {% for action in recommendation.actions %}
        <div class="flex items-center gap-3 p-3 bg-white border border-slate-100 rounded-xl text-xs text-slate-600 shadow-sm">
            <i data-lucide="chevron-right" class="w-4 h-4 text-indigo-400"></i> {{ action }}
        </div>
        {% endfor %}
    </div>
```

```
            </div>

            <button onclick="closeModal()" class="w-full py-4 bg-slate-800 hover:bg-slate-900
text-white font-bold rounded-2xl transition-all">
                CLOSE ASSESSMENT
            </button>
        </div>
    </div>
</div>
{% endif %}

<script>
    // Initialize Icons
    lucide.createIcons();

    function applyResultColors() {
        const card = document.getElementById('resultCard');
        if (!card) return;

        const resultColor = card.dataset.resultColor;
        if (!resultColor) return;

        card.style.borderColor = `${resultColor}22`;
        card.style.backgroundColor = `${resultColor}08`;
        card.style.borderLeftColor = resultColor;

        const iconBg = document.getElementById('resultIconBg');
        if (iconBg) iconBg.style.backgroundColor = resultColor;

        const title = document.getElementById('resultTitle');
```

```
        if (title) title.style.color = resultColor;

        const priority = document.getElementById('resultPriority');
        if (priority) priority.style.backgroundColor = resultColor;
    }

    applyResultColors();

    // Modal Close Function
    function closeModal() {
        const modal = document.getElementById('resultModal');
        if (modal) modal.style.display = 'none';
    }
  </script>
</body>
</html>
```

- ## Dataset details

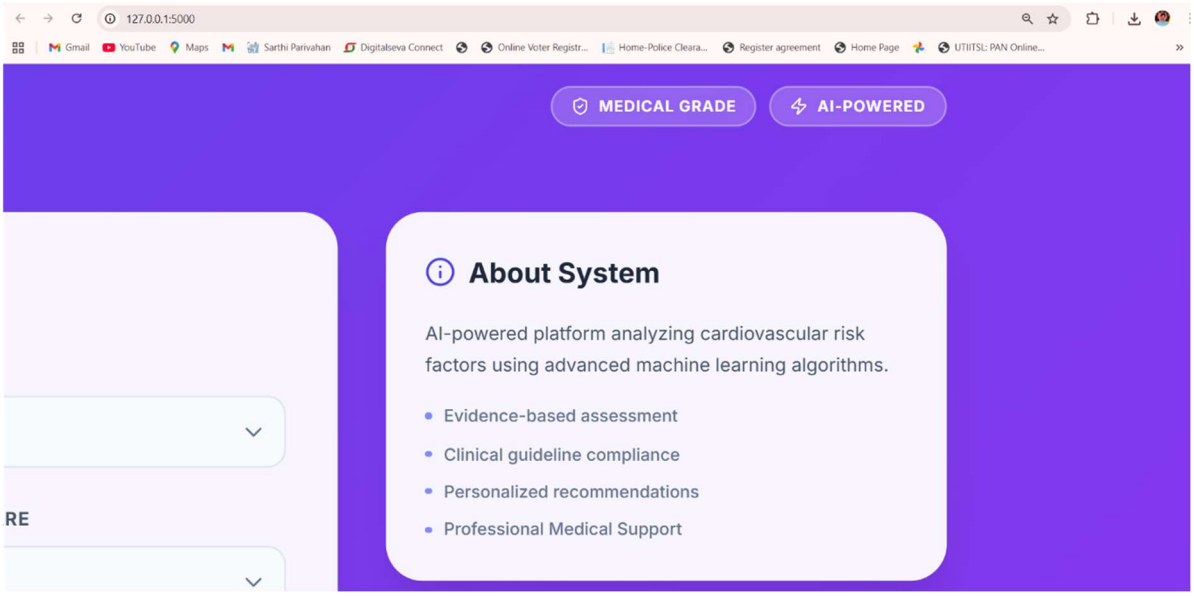
https://drive.google.com/file/d/1qYvKqg4w_w4blizSVqmLvwY25m7V7N3_/view?usp=sharing

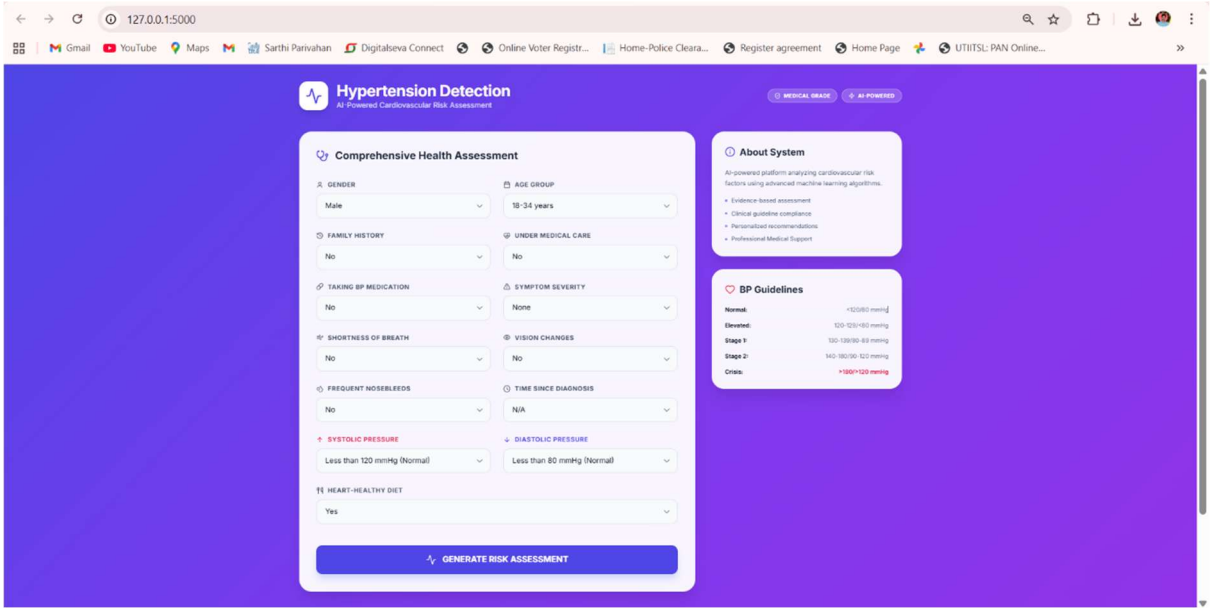
- ## Configuration files

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import joblib
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import accuracy_score, precision_score,
recall_score
```

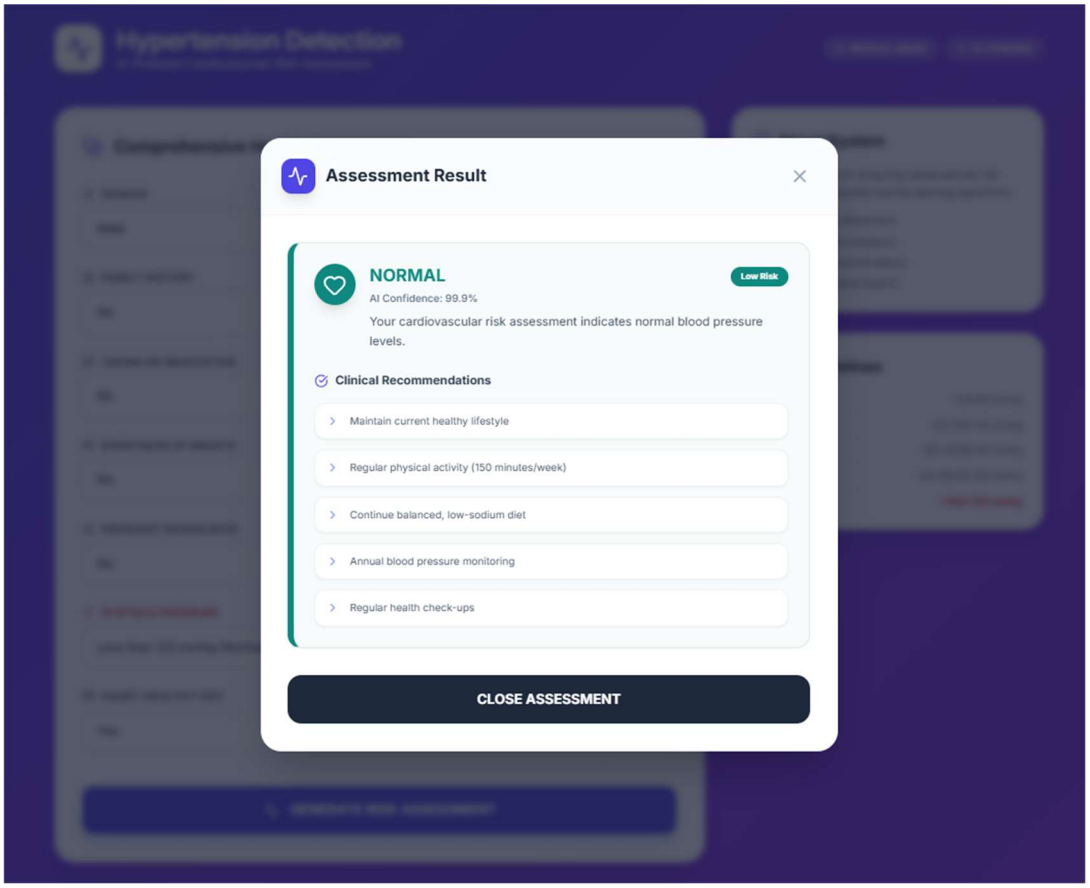
- GitHub repository link

- Screenshots of outputs

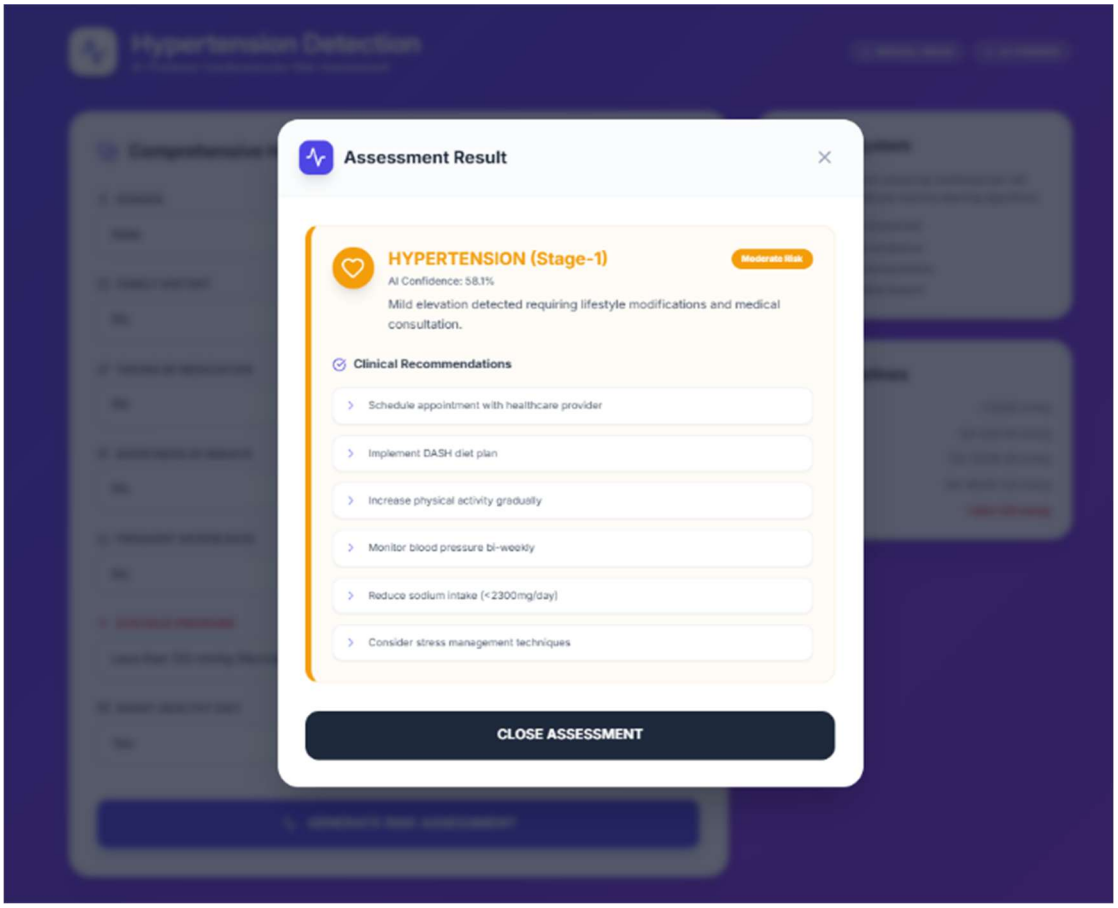
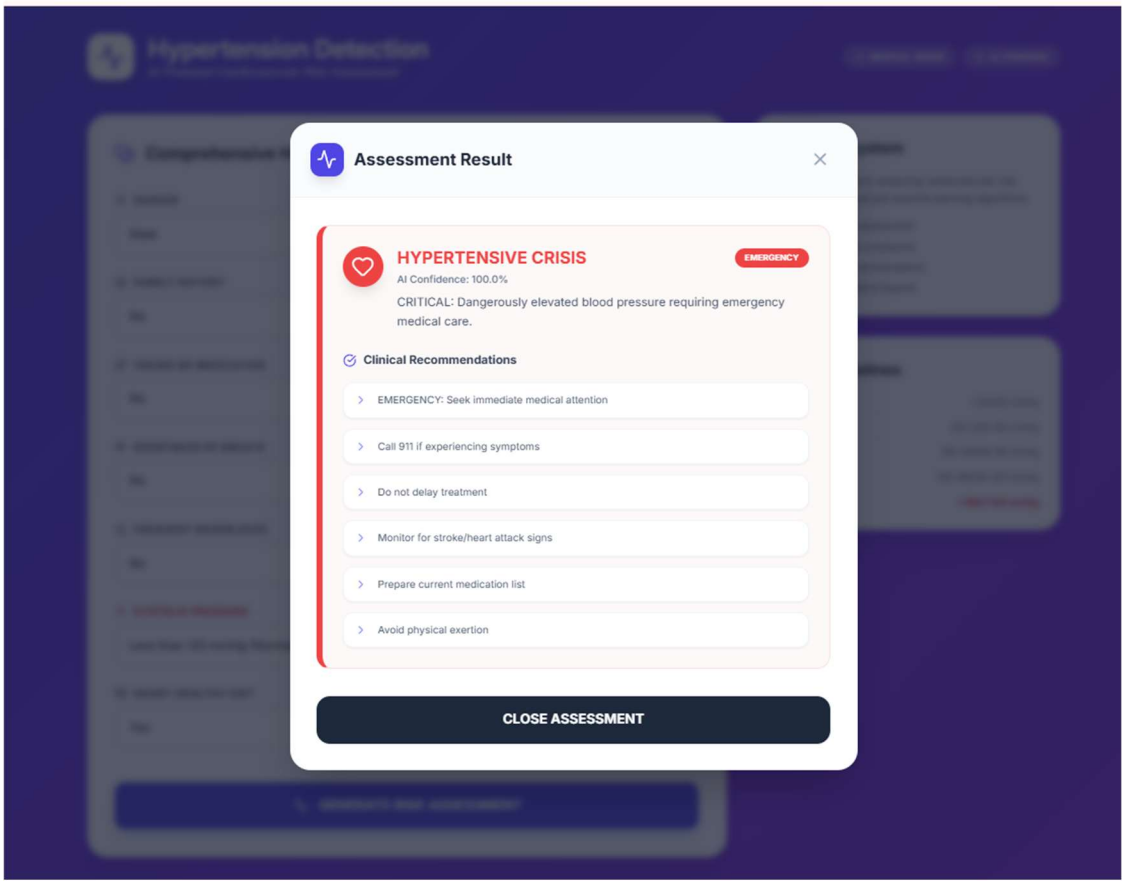




Result 1: Normal



Result 2: Moderate



Result 2: Emergency